

P R O G R E S S L O G

Mobile Charger

230 V AC → 5 V USB, from transformer to 3D-printed case

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01 Components We Used

Component	What it does
Transformer (230V→12V)	To step the mains voltage down to lower AC voltage.
4× rectifier diodes (1N4007, bridge configuration)	To convert the transformer's AC output into pulsating DC.
Smoothing capacitor C1	To smooth the pulsating DC into steadier voltage before it reaches the regulator. (Charges up during each rectifier pulse and discharges in between) <i>100 microfarad initially, ended up using a 2200 microfarad one instead</i>
Voltage regulator	To get the input voltage down from 15-17 V to 5V output voltage. 7805 voltage regulator (initially): linear voltage regulator LM2596 buck converter module (what we finally ended up using): switching voltage regulator <i>We tried to put the circuit together using both, ended up using the buck converter.</i>
USB-A female connector	—
Perfboard (5×7 cm)	—

02 Building the Circuit

Following the reference circuit

First, we wired everything according to the reference pdf:

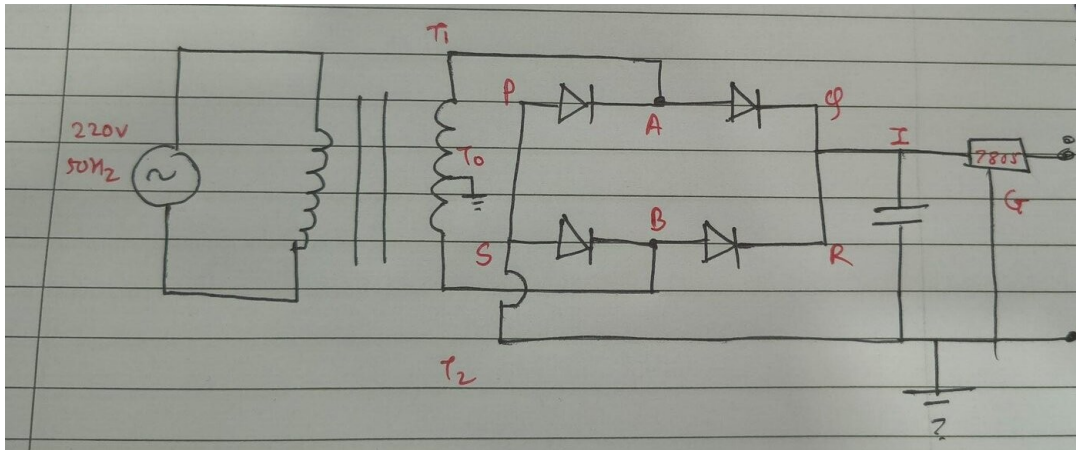


Figure 1. Reference circuit: transformer, bridge rectifier, smoothing capacitor and 7805.

Testing on the breadboard

We thought we'd first test it out on the breadboard before soldering:

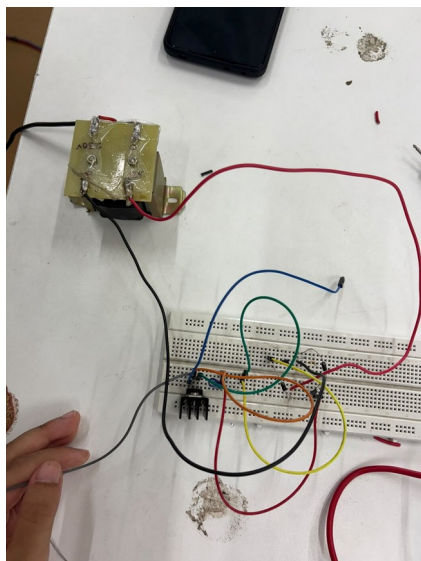


Figure 2. Circuit on the breadboard.

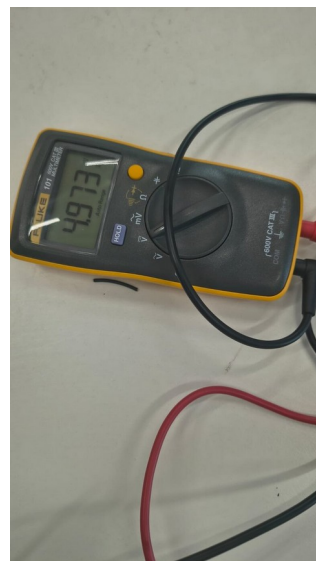


Figure 3. Multimeter showing 5V voltage.

Moving to the perfboard

Then we started setting up the circuit onto the perfboard:

Component mishaps

While soldering the middle pin of 7805 the outer one accidentally broke so we had to redo everything, desolder the pins we had already soldered, get a new voltage regulator and solder it again.

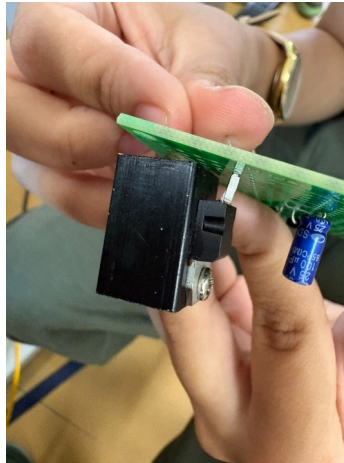


Figure 4. The 7805 on the perfboard.

Then we tried to check if the new 7805 worked by connecting the multimeter but of course it didn't work. But then after observing the circuit carefully we noticed that one of the diodes was inverted (again). Then we had to desolder that diode and fix it. But it finally worked (yay).

Soldering issues

The soldering iron we issued from Core Labs wasn't working properly. There was already a lot of solder on the tip, but the fresh solder wasn't wetting the tip properly and was instead forming droplets and falling off. We tried cleaning the tip by sanding it with the rough side of a nail cutter, since we didn't have sandpaper. For some reason, it still wasn't working, and we couldn't figure out why (So much ragebait.) but luckily we managed to get our hands on another one.

After soldering the circuit onto the perfboard, we checked it again and tested it with the multimeter first, thankfully, it worked.

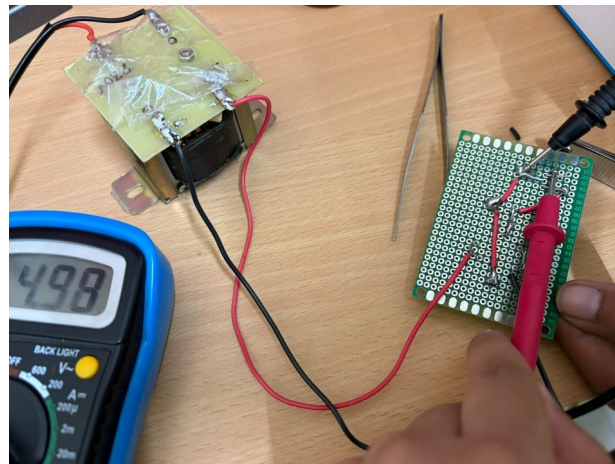


Figure 5. Testing the soldered perfboard with the multimeter.

Figuring out the USB connections and placement

This turned out to be a little more difficult than the rest of the circuit, to be honest. We weren't sure whether to place it above the perfboard and connect the wires through the holes and then solder it to the pins, or attach it underneath, which would make soldering a lot easier.

And also, do we attach it horizontally or vertically? If we attach it vertically, the pins are too short to go through the perfboard, and that would also make the 3D modelling a little awkward because where do

we put the hole for the usb? Attaching it horizontally seemed like a more viable option, and we ended up attaching it below the perfboard itself, because it made soldering easier.

We tried using cellophane, double sided tape, fevicol, feviquick and soldering it onto the perfboard as well but unfortunately we don't have access to a hot glue gun. I think that would have worked out better, since soldering the usb onto the perfboard was proving to be difficult. We ended up just sticking it onto the perfboard using feviquick and it seemed to be holding up well.



Figure 6. Charging a phone from the perfboard.

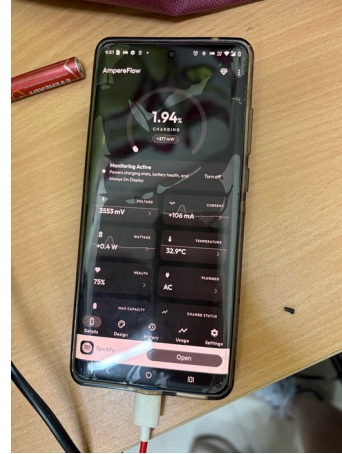


Figure 7. Charging stats with the 7805.

03 3D Printing the Case

We decided to use OpenSCAD, since we got the hang of it pretty easily.

Draft 1

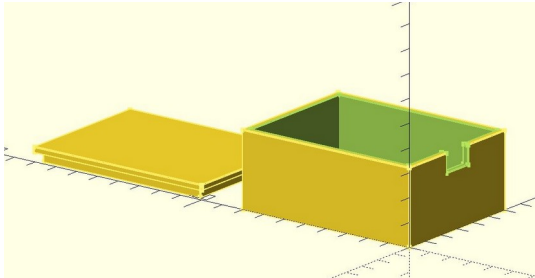


Figure 8. Draft 1: box with a lid.

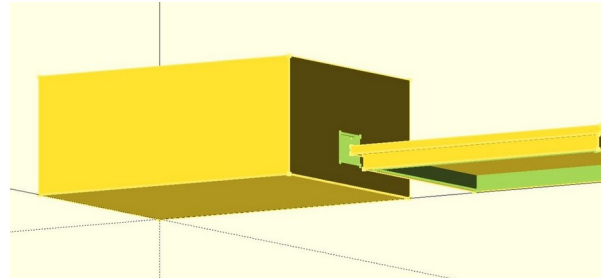


Figure 9. Draft 1: side view.

In our first draft, we decided to make a very simple, traditional box with a lid, with the transformer and the perfboard side by side. However, we eventually realised that the design wasn't very feasible, because if the transformer and the perfboard were side by side, it would make the charger very bulky, and we would have to make a partition in the middle which could get kind of messy. And we'd need a ton of supports for both the perfboard and the transformer.

Draft 2

We were kind of confused about how to fix the transformer to the bottom. If we screw it, we'll have to make the bottom super thick and also the holes of the transformer are oddly shaped, i dont know if we'll even be able to use a screw or do we super glue it to the bottom? Not sure if that'll work since the transformer is pretty heavy. We considered soldering the transformer to the bottom, but we finally decided on screwing it in from the bottom up since that seems like the only viable option.

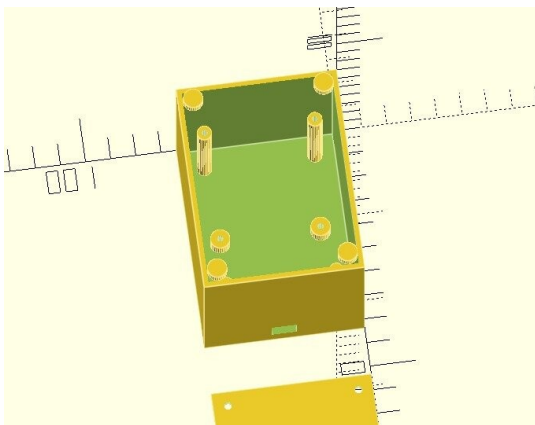


Figure 10. Draft 2: standoffs for the perfboard.

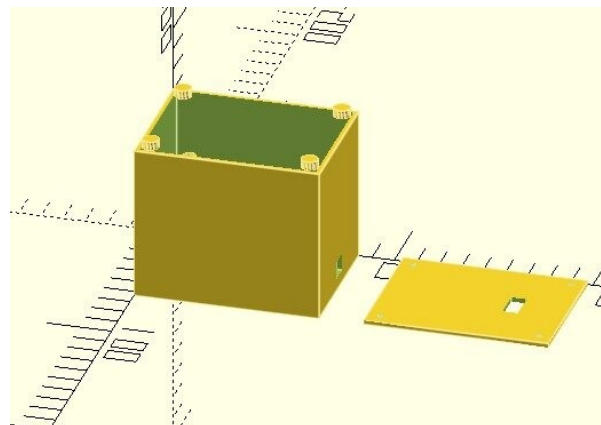


Figure 11. Draft 2: with the lid.

We thought we'd change the lid to a screw-on type one, to prevent things from falling out. First we thought we'd directly attach the perfboard to the top of the transformer but then we realised it might get heated up and the wires on the perfboard might get accidentally de-soldered so we added 4

supports to mount the perfboard on top of the transformer in such a way that they don't touch each other. The holes on the perfboard should be perfectly aligned with the holes on the standoffs if our measurements are correct, so we can screw the perfboard onto the stands. We still hadn't decided if we were doing a screw on lid or a sliding one, but we finally decided on a sliding one, because if we screw the lid on, we were going to have to make the lid even higher than the perfboard to make room for the screws which we didn't want to do since the charger is already really bulky.

Draft 3

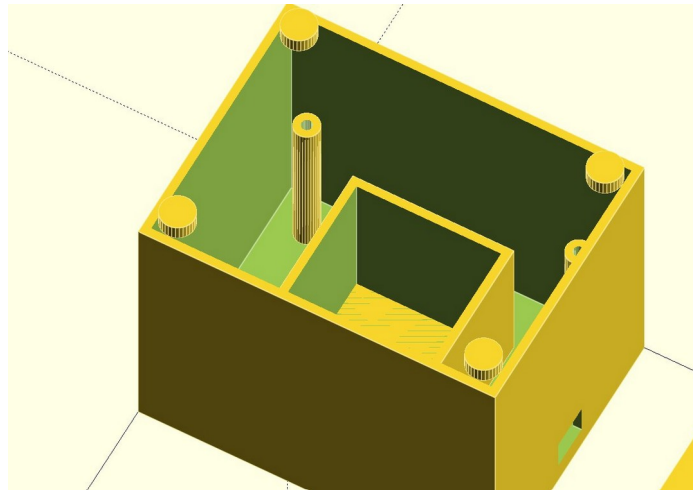


Figure 12. Draft 3: walls around the transformer.

We realised the screws might not be sufficient to stop the transformer from wobbling inside the case, first we thought of putting walls around the transformer, to stop it wobbling sideways, but for that we'd have to let go of the screws, and then there was nothing holding it in place vertically, unless we make a separate "lid" for the transformer, which wouldn't have worked either, since, the plate at the top with the wires connected, which has completely different dimensions from the transformer. The walls would only act as supports and what would actually hold it down would be the screws so we decided to change it again.

Draft 4

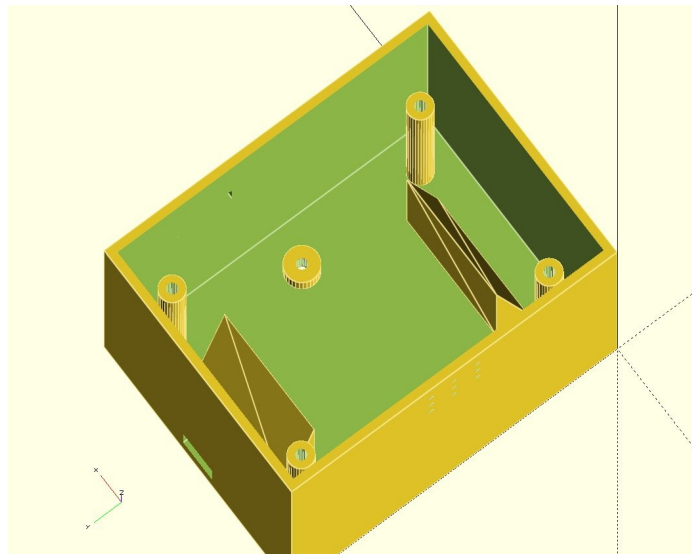


Figure 13. Draft 4: wedges to hold the transformer.

We decided on keeping the screws and adding wedges instead to hold the transformer in place. (It should work as long as we take the measurements correctly).

And for the wires exiting the transformer, do we add only 2 small holes for the wires or a rectangular cutout? The holes would be neater, but then we'd have to solder the wires after inserting the transformer inside the case, and im not sure how feasible that is, but if we go the cutout route, we'd have to make it pretty big to fit the plug as well, which would look pretty messy, but the hole would act as kind of a ventilator for the transformer, so we'll probably go with the rectangular cutout.

Draft 5

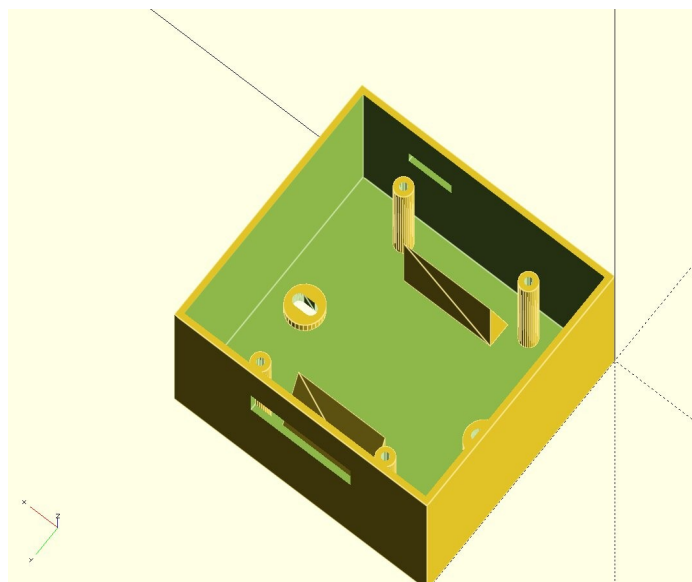


Figure 14. Draft 5: exact screw holes and ventilation.

We made a couple of changes, got the holes the exact dimensions of the transformer holes, at a height of 6mm, to give some leeway for the screw. And added ventilation holes at the bottom for the transformer.

04 Increasing the Output Power

Coming back to the charger

The problem

The only problem was, we were getting really low wattage while charging (around 0.5 W on average) even though it kept fluctuating between 0 and 5 W.

Figuring out how to increase the output power

Adding extra capacitors

We replicated the circuit again on a breadboard increasing the capacitance from 100 microfarad to 470 microfarad, and adding 0.1 microfarad ceramic capacitors at the input and output pins, but that still wasn't increasing the output by any significant margins. It did give the 7805 a noticeably steadier input voltage, but didn't fix the heat problem.

Why didn't adding capacitors work?

The 7805 wastes $(V_{in} - V_{out}) \times I$ as heat regardless of how smooth the input is, a more stable 17V input still has to drop 12V down to 5V, and that 12V drop across whatever current you're drawing still turns into heat. A bigger capacitor doesn't change that at all; it only affects the minimum voltage during dips, not the average voltage the regulator has to burn off, so that's probably why we didn't see any significant changes.

Adding a heat sink to the 7805?

We tried to get a heat sink, but we weren't able to get it from the lab and without a heatsink, we'd have hit the 7805's thermal ceiling well before reaching anywhere near 5W.

We'd also still be limited by the 7805's absolute maximum current rating (1A, less with poor cooling), so it caps out well below what current we actually required, so that wasn't solving the problem either.

Using a higher-current linear regulator?

(LM338/LM350, rated 3A)

It has a higher current ceiling than the 7805 but the same fundamental inefficiency as the 7805 — still a linear regulator, still wastes the voltage difference as heat. At higher current, the heat problem gets worse, not better, so we'd need an even bigger heatsink than the first way. Doesn't actually solve the core issue, just delays it.

Using a lower-voltage transformer

(e.g. 9V instead of 12V)

To reduce the voltage the regulator has to drop. Less voltage difference across the regulator means less wasted heat per amp, even with a linear regulator. Again, we don't really have access to another transformer, so we didn't look into this too much.

Adding a second regulator in parallel to split the current load

It would split current draw across two devices, each handling less individually. But the connections looked way too messy and complicated, and i dont think we could have fit it on the perfboard. Plus, I read that one regulator tends to hog most of the load, so that kind of defeats the purpose anyways.

Switching to a buck converter

(what we actually did — LM2596)

Switching regulation only dissipates a small fraction of the power as heat (typically 80–90%+ efficient), instead of burning off the entire voltage difference like a linear regulator. And thats what lets it supply more current from the same input without needing a large heatsink.

It was the only viable option that removed the actual root cause (linear regulation was wasting most of the power as heat) rather than working around it with bigger heatsinks. And it would be easier to use a readily available breakout module rather than something we'd need to design current-sharing circuitry for. The tradeoff is that switching regulators are electrically noisier (high-frequency switching creates ripples) whereas a linear regulator like the 7805 gives a quieter output.

Our decision

But for our project, we decided that the tradeoff clearly favours the LM2596 since a little bit of noise is worth getting a 5W output as compared to the 0.5W output we were getting with the 7805.

05 Starting Afresh with the LM2596

We decided to try our hands with the buck converter, we didn't want to risk desoldering our original circuit in case the new one didn't work so we got a new perfboard, diodes and usb and fiddled with that instead. Getting a buck converter in the first place proved to be a more difficult task than actually setting up the circuit.

We finally did get our hands on one and soldered everything, the connections were pretty similar to the original circuit, only a couple of minor changes.

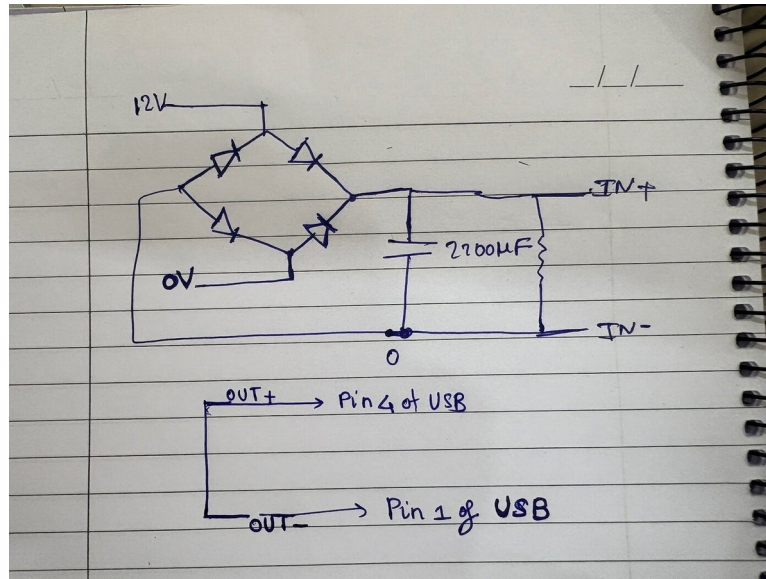


Figure 15. Circuit with the LM2596 and 2200 μ F capacitor.

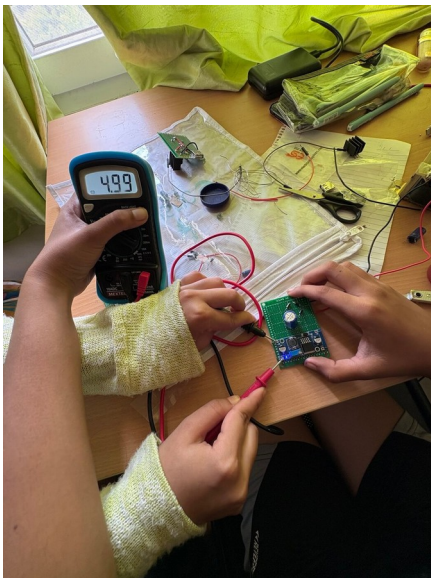


Figure 16. Output with the LM2596 on the multimeter.



Figure 17. Charging at 5.31 W with the LM2596.